

## Synchronous TMR Architecture with Single-Cycle FDI for Radiation-Prone FPGA DAQ Systems

Author: Faiza Khoso (Independent Research, 2026)

Repo: [github.com/faizakhoso1-prog/Resilient-FPGA-Muon-DAQ](https://github.com/faizakhoso1-prog/Resilient-FPGA-Muon-DAQ)

### Problem

Radiation-induced Single Event Upsets (SEUs) flip FPGA state mid-operation, corrupting control cores and counters in unshielded, safety-critical DAQ pipelines (e.g., cosmic-ray muon detection).

### Solution

Independently engineered a fully synchronous Triple Modular Redundancy (TMR) control core in VHDL on Xilinx Artix-7 (xc7a100tcsg324-1, Nexys 4 DDR). Three parallel hardware cores are masked by a glitch-free registered majority voter. A single-cycle Fault Detection & Isolation (FDI) telemetry engine flags the corrupted core (01=A, 10=B, 11=C) and outputs handshake flags (sig\_fault\_det, sig\_faulty\_core) for ICAP scrubbers or DFX controllers.

### Key Results

- Timing closure: 100 MHz post-implementation (WPWS = 4.500 ns, zero failing endpoints).
- SEU validation: Vivado behavioral simulation with runtime SEU injection — Core A corrupted at 250 ns; voter held correct state; fault flagged next clock edge; system continued uninterrupted; Core A auto-resynchronized at 450 ns. Zero data loss and zero downtime.
- Ultra-low footprint: 33 mW core logic (7 LUTs, 5 FFs).
- Registered majority voter:  $\text{Voted\_State} = (A \text{ AND } B) \text{ OR } (B \text{ AND } C) \text{ OR } (A \text{ AND } C)$ ; eliminates routing hazards and transient clock-skew glitches.

### Integration & Impact

TMR is embedded into 16-bit TDC counters in a Zynq-7000 DAQ (PL high-speed acquisition → ARM Cortex-A9 PS via AXI4). This protects timestamping/tracking data from bit-flips and enables real-time health monitoring via FDI telemetry. Manuscript in preparation (hardened electronics venues).

### Reproducibility

GitHub includes Vivado timing summary, SEU-injection logs, testbenches, and documentation assets.